

Resources: HSMA 6 Recordings



HSMA

2B: SimPy for Discrete Event Simulation - Part 1

3 hours

SESSION LEAD: DR DANIEL CHALK

In this session, we look at how we can use the SimPy Python package (which is built around generator functions) to build Discrete Event Simulations in Python.

We start with a very simple model in which patients arrive, are seen by a nurse, and then leave, before building up to multi-step models and models with branching pathways. We then explore how to record results from our simulation and explore the impact of changing the level of resources available or the distribution of inter-arrival times and activity durations.

This content is also covered in part 2 of the HSMA little book of DES.

This session makes use of the [des](#) environment

📄 Slides

📺 Video

📁 Repository

🖥️ Open in Github
Codespaces

📖 Book

📺 Exercise Solution Video
- Part 1

📺 Exercise Solution Video
- Part 2

2C: SimPy for Discrete Event Simulation - Part 2

3 hours

SESSION LEAD: DR DANIEL CHALK

In this session, we look at some more advanced DES concepts and how we can incorporate them in our SimPy models.

The full recordings from HSMA 6's DES module can be found at hsma.co.uk/des

All of the other HSMA 6 content can be found at hsma.co.uk/modules

In this session, we've taught our new, improved approach to DES model structure - but these recordings and exercises still provide a great starting point.

Resources: Little Book of DES



HSMA



Licencing, Attribution and Acknowledgements

Part 1 - DES Concepts and Design

- 1 Introduction to DES Concepts
- 2 Exercise - Designing a DES

Part 2 - Your First SimPy Model

- 3 An Introduction to SimPy
- 4 The Recommended Structure for DES Models
- 5 An Example SimPy Model
- 6 Adding Multiple Activities
- 7 Adding Branching Paths
- 8 Exercise - Building Your First Model

Part 3 - Extending Your Basic Model

- 9 Warm Up Periods
- 10 Priority-Based Queueing
- 11 Modelling Resource Unavailability
- 12 Using Different Distributions for Different Processes

```
# Simulation meta parameters
sim_duration = 2880
warm_up_period = 1440 ##NEW - this will be in addition to the sim_duration
number_of_runs = 100
```

Tip

If you find it easier to keep track of, you could define your warm-up like this instead.

```
results_collection_period = 2880
warm_up_period = 1440
total_sim_duration = results_collection_period + warm_up_period
```

9.3.2 The patient class

Our patient class is unchanged.

9.3.3 The model class

In the model class, the 'attend_clinic' method changes.

We look at the current elapsed simulation time with the attribute `self.env.now`

Then, whenever a patient attends the clinic and is using a nurse resource, we check whether the current simulation time is later than the number of time units we've set as our warm-up.

9.3.3.1 The attend_clinic method

```
# Generator function representing pathway for patients attending the
# clinic.
def attend_clinic(self, patient):
    # Nurse consultation activity
    start_q_nurse = self.env.now

    with self.nurse.request() as req:
        yield req
```

Table of contents

- 9.1 How long should a warm-up period be?
- 9.2 Implementing the warm-up period
- 9.3 Coding the model
 - 9.3.1 The g class
 - 9.3.2 The patient class
 - 9.3.3 The model class
 - 9.3.4 The trial class
- 9.4 The impact of the warm-up period

Our free eBook at des.hsma.co.uk covers all content from the HSMA 6 DES module, as well as significantly expanding it with more advanced concepts.

Again, this uses the older approach to writing HSMA models, but it still provides a great resource for this stuff!

Built something?



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If you continue your DES journey from today and build a model, we would love to hear about it!

Please email us on [**penchord@exeter.ac.uk**](mailto:penchord@exeter.ac.uk) or tag us in a message on the NHS-OA Slack.

Resources: DES RAP Book



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DES RAP

Step-by-step guide

Examples

Frameworks ▾

Intros ▾

Impact

GitHub

Choose your language:

Python

R

Welcome to our step-by-step guide to DES RAP

PAGE LAST MODIFIED
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Introduction

Setup ▾

Version control

Environments

Structuring as a package

Code organisation

Conceptual modelling

Model inputs ▾

Input modelling

Input data management

Parameters from script

Parameters from file

Parameter validation

Model building ▾

Randomness

Entity generation

Entity processing

Logging

Output analysis ▾

How to use this guide

Work through each page **in order**. Follow along with the code examples as you go. You'll gradually build up a working model and your skills as you progress.

Each section builds on the previous one. You can:

- Follow the explanations and instructions
- Copy and run the provided code
- Complete exercises at the end to test your understanding

It will help you meet two important reproducibility standards:

NHS Levels of RAP

STARS reproducibility recommendations

On this page

[How to use this guide](#)

[Before you get started...](#)

[Example models](#)

[Getting help](#)

[Reporting problems](#)

Produced by our ARC colleague Amy Heather, this book is a thorough guide to reproducible DES modelling in both Python and R.

pythonhealthdatascience.github.io/des_rap_book

Resources: Atlas



HSMA

The Healthcare Services Analytics & Decision Support Atlas collates projects from the NHS and related organisations.

The DES page collates a range of resources and models:

atlas.hsma.co.uk/techniques/des/

Resources: Vidigi Package



HSMA

[vidigi](#) [Getting Started](#) [Customising Animations](#) [Feature Breakdowns](#) [API Reference](#) [Example Models](#) [Migrating to 2.0.0](#) [Changelog](#) [About](#) ▾



Examples Gallery

Explore what's possible with vidigi through these hands-on examples.

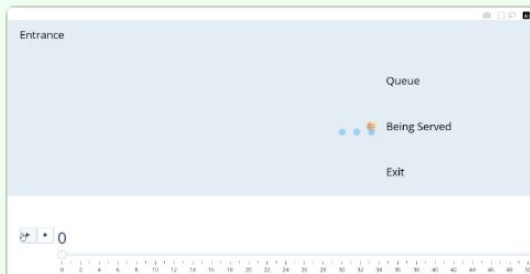
SimPy Examples

A collection of models built using the SimPy framework.



A very simple example with one server

The simplest possible starting point - a single-step SimPy model with just one server. This example walks through using vidigi's EventLogger helper class to capture arrivals, queueing and resource use with the minimum of code changes to an existing model.



A slightly more complex example with multiple servers

Building on the single-server example, this model introduces multiple servers, so we need a way to track which specific resource each entity is using. Here we bring in vidigi's VidigiStore class, a drop-in replacement for SimPy's Resource

On this page

[SimPy Examples](#)

[Ciw Examples](#)

[Salabim Examples](#)

[Agent Based Simulation \(ABS\) Examples](#)

[Advanced Examples](#)

[Legacy Examples](#)

If you are looking to add vidigi to your own models, check out the documentation to find a 'getting started' guide, extensive examples, and more.

hsma-tools.github.io/vidigi/

ReModel Club: Read, Run, Reuse Open Simulation Models



The ReModel club is a new club run by the strategy unit, focussing on **Running** and **Reusing** healthcare simulation models.

Running ~monthly for 1 hour, their goals are to “empower people within the NHS to:

- run open source simulation models on their own machines
- critique/assess/evaluate simulation models with their pros and cons
- repurpose existing simulation models for their own use cases”

Sign up here:

the-strategy-unit.github.io/ReModel-docs/join-us/#register-for-remodel

The sessions are free to attend, but spaces will be capped at 15 places per session on a first-come, first-served basis; NHS analysts will receive priority.

Sign up for HSMA 7



HSMA

We are hoping to launch HSMA 7 in late Spring/early Summer 2027.

HSMA 7 will feature our biggest rewrite yet, with the DES module expanding from 9 hours to **over 30 hours of content**, covering DES modelling in Python in more detail than ever before.

Find out more by visiting [**hsma.co.uk/hsma7**](https://hsma.co.uk/hsma7)

Want to hear when applications open?

Sign up to our mailing list by emailing
[**penchord@exeter.ac.uk**](mailto:penchord@exeter.ac.uk)

Follow us on LinkedIn at
[**linkedin.com/company/hsma-programme/**](https://linkedin.com/company/hsma-programme/)

What else will you learn on HSMA 7?



In HSMA 7, we are doubling down on our two most impactful areas.

“Where is demand, and where should I put my services?”

Geographic Modelling

In this module, students learn key geographic principles and how to create high-quality maps.

They are also introduced to advanced operational research techniques for solving problems around where to place services in the face of competing priorities like equity of access, costs, and net zero goals.



“I want to identify patients to target with interventions”

Machine Learning

In this module, students learn the key principles of supervised machine learning, allowing them to train machine learning models designed to support operational improvements to services through better targeting of interventions.

They are also introduced to issues around model bias, ethics, legal issues, and model explainability.



What else will you learn on HSMA 7?



Web Application Development

Students will learn to use the Streamlit app framework, allowing them to create engaging interactive web apps to bring powerful techniques like simulation into the hands of non-coders



Interactive Reporting

Students will learn Quarto, a publishing system that allows them to build parameterised data-driven static and interactive reports



Data Visualisation

Students will learn to generate engaging static and interactive charts and maps in Python with industry-standard packages



Automation

Students can combine these techniques to automate and improve regular reporting

How is HSMA Structured?



PHASE 1: Learn

6 months training, 1 day a week



PHASE 2: Apply

9 months on a real project for your organisation

1 day a week expectation, with guidance and support



PHASE 3: Implement

From month 15 onwards, receive support for implementation and impact gathering



Need More Convincing?



HSMA

“My HSMA journey has been a transformative experience, opening me up to new approaches to analytics in the NHS.”

“HSMA is an absolutely amazing programme that’s taken me from zero coding knowledge to inspiring me to want to go off and explore on my own.”

“I can honestly say it is the best training I’ve ever participated in. For the first time in a long time I feel really enthused about my work.”